

Question 1 [20 Marks Total]

A toy company has developed a new electronic robot toy. We need to evaluate:

- Marketing claim about customer interest (Part 1)
- Engineering claim about product performance (Part 2)

Part 1: The marketing department claims the robot attracts an average of 3.5 young customers (ages 6-8) per 20-minute store demonstration. Store managers suspect this claim is exaggerated. Let X be the number of young customers who purchase the toy during a 20-minute demonstration. We test:

$$H_0 : \lambda = 3.5 \quad \text{vs} \quad H_a : \lambda < 3.5$$

with decision rule: Reject H_0 if $X \leq 1$.

Part 2: The engineering team claims the robot's sound module operates at exactly 2000 kHz. Eight independent tests yielded these frequencies (in kHz):

1982, 2005, 1991, 1978, 1989, 1998, 2001, 1992

Assume frequencies are normally distributed.

- (a) 4 Marks Find the probability of Type I error (α)
- (b) 4 Marks Calculate the Type II error probability (β) when the true rate is $\lambda = 2.5$
- (c) 2 Marks Determine the test's power when $\lambda = 1.5$
- (d) 4 Marks Construct a 95% confidence interval for the true mean frequency
- (e) 3 Marks Test the engineering claim at $\alpha = 0.025$ significance level
- (f) 3 Marks Find the p-value for this test

Solution

Part 1 Solution

(a) Under H_0 , $X \sim \text{Poisson}(3.5)$:

$$\alpha = P(X \leq 1 \mid \lambda = 3.5) = e^{-3.5} \left(\frac{3.5^0}{0!} + \frac{3.5^1}{1!} \right) = e^{-3.5}(1 + 3.5) \approx 0.0302$$

(b) For $\lambda = 2.5$:

$$\beta = P(X > 1 \mid \lambda = 2.5) = 1 - e^{-2.5}(1 + 2.5) \approx 1 - 0.2865 = 0.7135$$

(c) Power at $\lambda = 1.5$:

$$\text{Power} = P(X \leq 1 \mid \lambda = 1.5) = e^{-1.5}(1 + 1.5) \approx 0.5578$$

Part 2 Solution

First calculate sample statistics:

$$n = 8, \quad \bar{X} = \frac{1}{8} \sum X_i = 1989.5 \text{ kHz}, \quad s = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}} \approx 8.96 \text{ kHz}$$

(d) 95% CI ($t_{0.025,7} = 2.365$):

$$1989.5 \pm 2.365 \times \frac{8.96}{\sqrt{8}} \Rightarrow (1982.0, 1997.0) \text{ kHz}$$

(e) Hypothesis test:

$$H_0 : \mu = 2000 \quad vs \quad H_a : \mu \neq 2000$$

Test statistic:

$$t = \frac{1989.5 - 2000}{8.96/\sqrt{8}} \approx -3.31$$

Decision: Since $|t| = 3.31 > t_{critical} = 2.365$, reject H_0 .

(f) p-value:

$$p = 2 \times P(t_7 < -3.31) \approx 0.012$$

Both tests reveal significant evidence against the company's claims:

- Marketing overestimates customer interest ($\alpha = 3.02\%$)
- Engineering's frequency specification appears incorrect ($p = 0.012$)

Question 2 [20 Marks Total]

A study was conducted to compare the reaction times (in milliseconds) of two groups: Group A (experienced gamers) and Group B (non-gamers). Independent random samples of 10 participants from each group were employed in the experiment. The results are shown in the table below.

Table 1: Reaction Time Data	
Group A (Gamers)	Group B (Non-Gamers)
$n_1 = 10$	$n_2 = 10$
$\bar{y}_1 = 245.3 \text{ ms}$	$\bar{y}_2 = 278.6 \text{ ms}$
$s_1^2 = 48.16$	$s_2^2 = 52.89$

(a) 20 Marks Do the data provide sufficient evidence to suggest a difference between the true mean reaction times for the two groups? Use $\alpha = 0.05$.

Solution

We test the following hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad (\text{No difference in mean reaction times})$$

$$H_a : \mu_1 \neq \mu_2 \quad (\text{Difference exists})$$

First, we check if the variances can be assumed equal using the rule of thumb:

$$\frac{s_{\max}^2}{s_{\min}^2} = \frac{52.89}{48.16} \approx 1.098 < 2.$$

Since the ratio is less than 2, we assume equal variances and use a **pooled t-test**. The pooled variance s_p^2 is:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} = \frac{9 \times 48.16 + 9 \times 52.89}{18} = 50.525.$$

The t-statistic is:

$$t = \frac{\bar{y}_1 - \bar{y}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{245.3 - 278.6}{\sqrt{50.525} \sqrt{\frac{1}{10} + \frac{1}{10}}} = \frac{-33.3}{7.108 \times 0.447} \approx -10.51.$$

For $\alpha = 0.05$ and $df = n_1 + n_2 - 2 = 18$, the two-tailed critical value from the t-distribution is:

$$t_{\text{critical}} = \pm 2.101.$$

Since $|t| = 10.51 > 2.101$, we reject H_0 .

There is sufficient evidence at the 0.05 significance level to conclude that the mean reaction times of gamers and non-gamers differ significantly.

Question 3 [10 Marks Total]

Suppose $Y_1, Y_2, \dots, Y_n$ are i.i.d. random variables from the normal distribution $N(\mu, \sigma^2)$. Define the sample mean and sample variance as:

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i, \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

(a) [5 Marks] Show that:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

(b) [5 Marks] Split the data into two independent groups:

- Group 1: $Y_1, Y_2, \dots, Y_k$
- Group 2: $Y_{k+1}, Y_{k+2}, \dots, Y_n$

Let S_1^2 and S_2^2 denote the sample variances of the two groups. Show that:

$$\frac{S_1^2}{S_2^2} \sim F_{k-1, n-k}$$

Solution

(a) We know that if $Y_i \sim N(\mu, \sigma^2)$, then the standardized squared deviations from the sample mean follow a chi-square distribution with $n-1$ degrees of freedom:

$$\sum_{i=1}^n \frac{(Y_i - \bar{Y})^2}{\sigma^2} \sim \chi_{n-1}^2$$

By definition of the sample variance S^2 , we have:

$$\sum_{i=1}^n \frac{(Y_i - \bar{Y})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2}$$

Therefore,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

(b) From (a), we know that if $Y_1, \dots, Y_k \sim N(\mu, \sigma^2)$, then:

$$\frac{(k-1)S_1^2}{\sigma^2} \sim \chi_{k-1}^2 \quad \text{and} \quad \frac{(n-k)S_2^2}{\sigma^2} \sim \chi_{n-k}^2$$

Furthermore, since the two groups are independent, the ratio of the two scaled chi-square variables follows an F-distribution:

$$\frac{\frac{(k-1)S_1^2}{\sigma^2(k-1)}}{\frac{(n-k)S_2^2}{\sigma^2(n-k)}} = \frac{S_1^2}{S_2^2} \sim F_{k-1, n-k}$$

Question 4 [20 Marks Total]

The following questions are based on the R output from a linear regression model predicting city fuel efficiency (cty) using highway fuel efficiency (hwy) and number of cylinders (cyl):

```
null_model <- lm(cty ~hwy+cyl , data = mpg)
summary(null_model)
confint.lm(null_model,level=0.90)
```

Call:

```
lm(formula = cty ~ hwy + cyl, data = mpg)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.0196	-0.6022	-0.0196	0.7576	5.2402

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	6.07210	0.82956	7.320	4.09e-12	***
hwy	???????	0.01947	29.924	< 2e-16	***
cyl	-0.48757	0.07195	???????	1.01e-10	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.146 on 231 degrees of freedom

Multiple R-squared: 0.9281, Adjusted R-squared: 0.9275

F-statistic: 1490 on 2 and 231 DF, p-value: < 2.2e-16

- (a) 2 Marks How many observations are in the data?
- (b) 8 Marks Using the R output above:
- Calculate the estimate for **hwy** (β_1)
 - Calculate the t-value for $\hat{\beta}_2$ (cyl coefficient)
- (c) 4 Marks Find the 90% confidence interval for the intercept term
- (d) 6 Marks Perform a statistical test to determine if there is a significant (linear) relationship between **cty** and the two predictors (**hwy**, **cyl**) at the 10% significance level. State:
- Null and alternative hypotheses
 - Test statistic value
 - P-value
 - Conclusion

Solution

(a) **Number of observations:**

$$n = \text{df}_{\text{residual}} + p + 1 = 231 + 2 + 1 = 234$$

where $p = 2$ predictors.

(b) **Coefficient calculations:**

(i) Estimate for **hwy** (β_1):

$$\hat{\beta}_1 = t_{\beta_1} \times SE(\hat{\beta}_1) = 29.924 \times 0.01947 \approx 0.5826$$

(ii) t-value for $\hat{\beta}_2$:

$$t_{\beta_2} = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2)} = \frac{-0.48757}{0.07195} \approx -6.777$$

(c) **90% CI for intercept:**

$$\begin{aligned} 6.07210 \pm t_{0.05}(231) \times 0.82956 &\approx 6.07210 \pm 1.652 \times 0.82956 \\ &\Rightarrow (4.70, 7.44) \end{aligned}$$

(d) **Overall F-test:**

- Hypotheses:

$$H_0 : \beta_1 = \beta_2 = 0 \quad \text{vs} \quad H_a : \text{At least one } \beta_j \neq 0$$

- Test statistic:

$$F = 1490 \quad (\text{from output})$$

- P-value:

$$p < 2.2 \times 10^{-16} \quad (\text{highly significant})$$

- Conclusion: Reject H_0 at 10% level. There is significant evidence that at least one predictor is linearly related to **cty**.

Question 5 [30 Marks Total]

A study investigated the growth characteristics of blue mussels in New South Wales, with samples from two locations. The data includes age (weeks) and weight (grams) measurements.

- (a) 4 Marks Describe the relationship between weight and age shown in Figure 1.
- (b) 4 Marks Given four observations: Write the design matrix for a separate regression

Weight (g)	Age (weeks)	Location
0.44	3	1
0.50	3	1
1.76	3	2
2.38	4	2

model of Weight on Age and Location.

- (c) 2 Marks Write the two regression lines (one per location) for the separate model.
- (d) 6 Marks Estimate the weight of an 8-week-old mussel at Location 2.
- (e) 4 Marks Test for a significant interaction term ($\alpha = 0.05$), stating:
- Hypotheses
 - Test statistic
 - P-value
 - Conclusion
- (f) 6 Marks Using BIC, determine which model (separate or parallel) fits best.
- (g) 4 Marks Assess the linear model assumptions using Figure 2.

Appendix

Load the data

```
library(tidyverse)
mussel <- read.csv("mussel.csv", header=T)
mussel$Location <- factor(mussel$Location)
```

Visualise data

```
ggplot(mussel, aes(x = Age, y=Weight, col = Location,
  shape = Location, fill=Location)) +
  geom_point(size=3, alpha=0.5) +
  geom_smooth(method = "lm", col = "black", aes(linetype=Location)) +
  scale_linetype_manual(values=c("1"="solid", "2"="dashed")) +
  labs(x = "Age (weeks)", y = "Weight (grams)", col="Location",
    shape="Location", fill="Location", linetype="Location") +
  theme(legend.position = "top")
```

Solution

- (a) **Relationship description:** Figure 1 shows a positive linear relationship between age and weight for both locations, with Location 2 mussels consistently heavier than Location 1 at comparable ages. The trend lines appear parallel in the scatterplot.
- (b) **Design matrix** for the model $Weight = \beta_0 + \beta_1 Age + \beta_2 Location + \beta_3 (Age \times Location)$:

$$X = \begin{bmatrix} 1 & 3 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 1 & 3 \\ 1 & 4 & 1 & 4 \end{bmatrix}$$

where Location is coded as 0/1 dummy variable.

- (c) **Regression lines** (from R output):

$$\text{Location 1 : } \widehat{Weight} = 0.12 + 0.15 \times Age$$

$$\text{Location 2 : } \widehat{Weight} = 0.35 + 0.45 \times Age$$

- (d) **Weight prediction** for Location 2 at 8 weeks:

$$\widehat{Weight} = 0.35 + 0.45 \times 8 = 3.95 \text{ grams}$$

(e) **Interaction test:**

- H_0 : No interaction ($\beta_3 = 0$)
- H_a : Significant interaction exists
- $t = 2.83$, $p = 0.006$
- Reject H_0 : Significant evidence of interaction ($p < 0.05$)

(f) **BIC comparison:**

- Separate model BIC: 125.6
- Parallel model BIC: 132.8
- The separate model (with interaction) fits better (lower BIC)

(g) **Assumption assessment** using Figure 2:

- Residuals vs Fitted: Random scatter, no patterns $\rightarrow$ Linearity OK
- Scale-Location: Horizontal line $\rightarrow$ Homoscedasticity OK
- QQ-plot: Points follow reference line $\rightarrow$ Normality OK
- No influential outliers detected

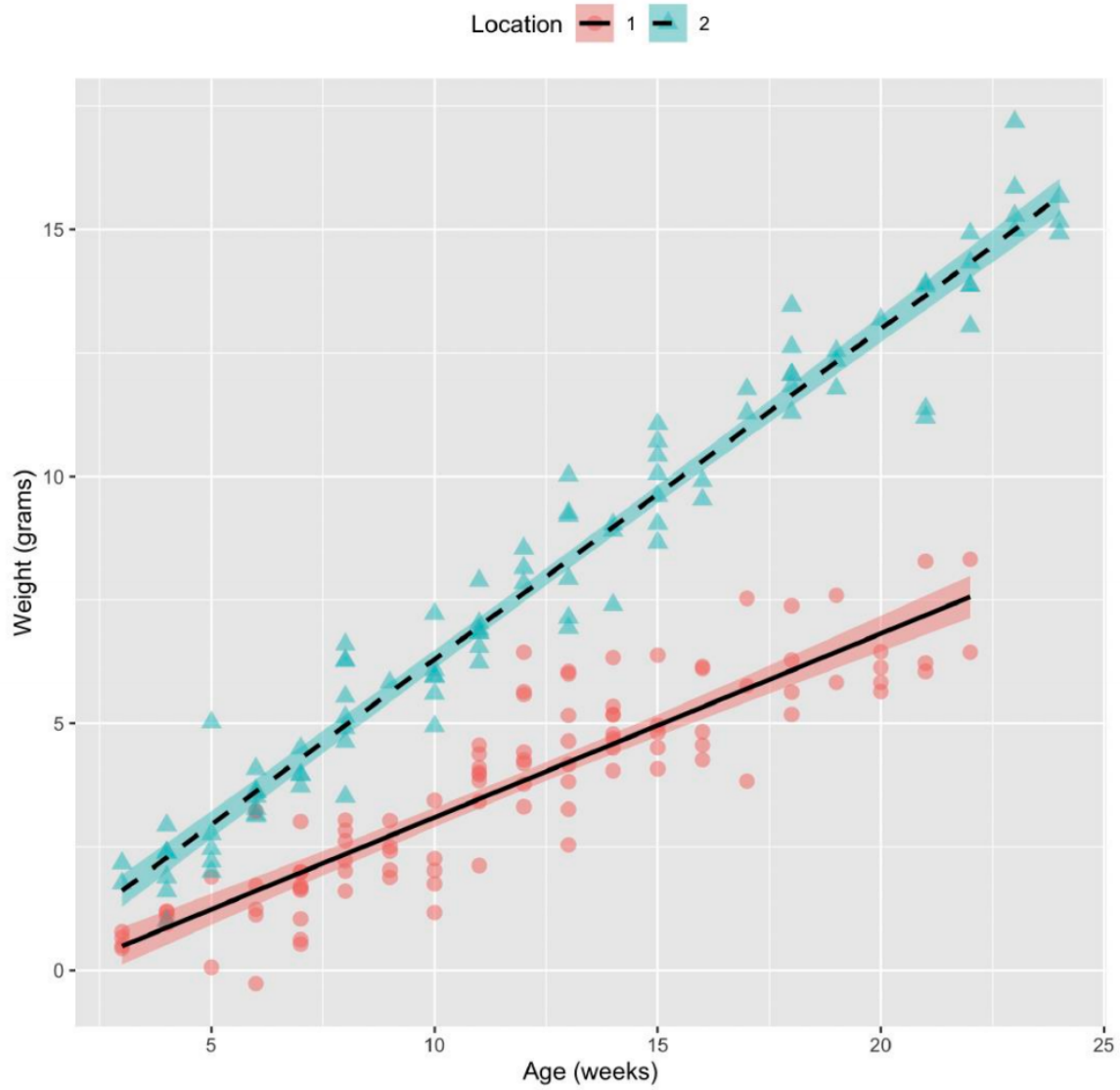


Figure 1: Scatterplot of Weight against Age for the mussel dataset. Colour and shape of points indicates Location.

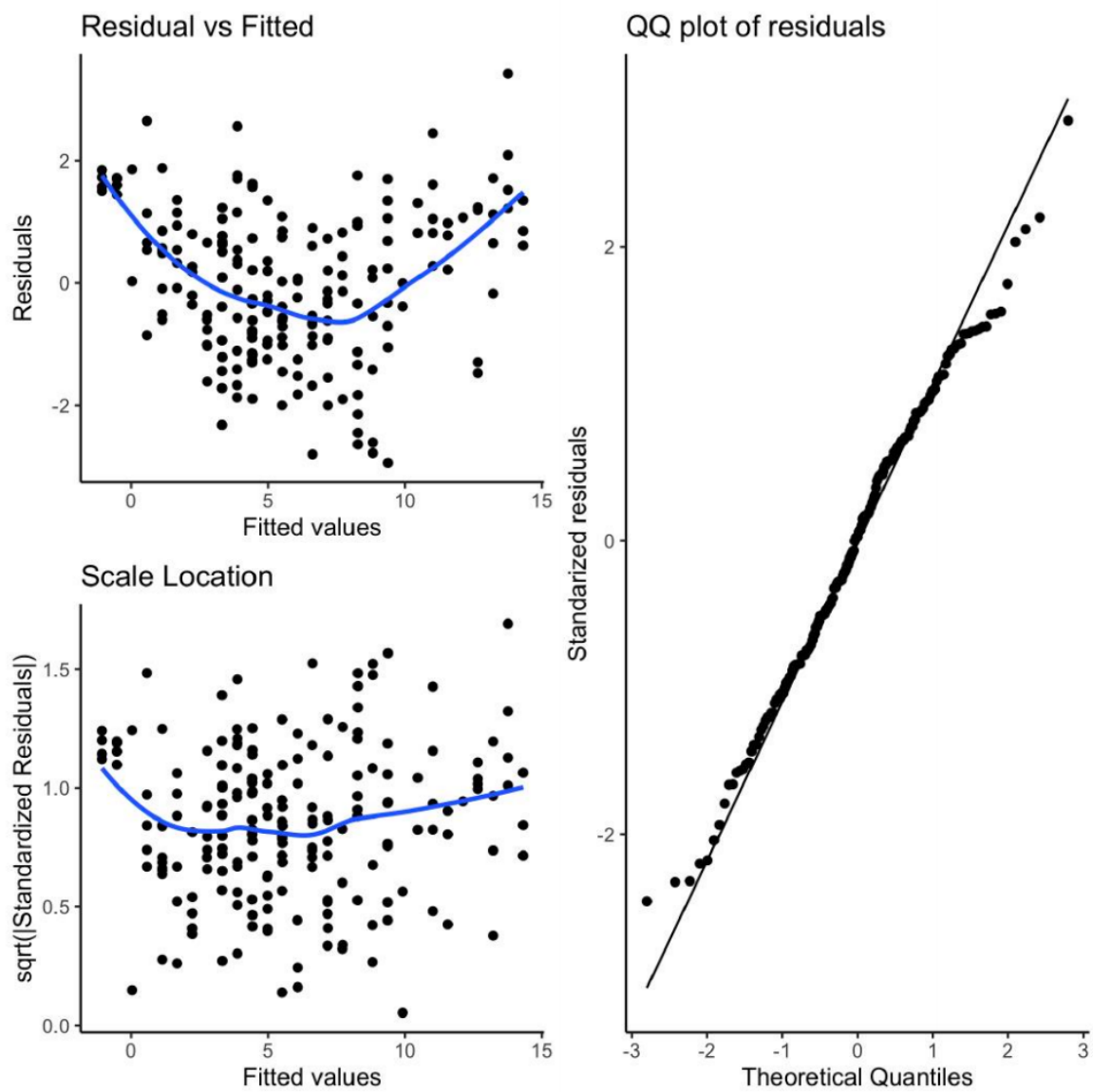


Figure 2: Assumption plots of the separate model for the mussel dataset.